```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
df = pd.read_csv ("/content/winedataset.csv")
```

```
print(df.head())
```

```
   Wine  Alcohol  Malic.acid   Ash  ...  Color.int   Hue    OD  Proline
0     1    14.23        1.71  2.43  ...       5.64  1.04  3.92     1065
1     1    13.20        1.78  2.14  ...       4.38  1.05  3.40     1050
2     1    13.16        2.36  2.67  ...       5.68  1.03  3.17     1185
3     1    14.37        1.95  2.50  ...       7.80  0.86  3.45     1480
4     1    13.24        2.59  2.87  ...       4.32  1.04  2.93      735

[5 rows x 14 columns]
```

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Wine                  178 non-null    int64
 1   Alcohol               178 non-null    float64
 2   Malic.acid            178 non-null    float64
 3   Ash                   178 non-null    float64
 4   Acl                   178 non-null    float64
 5   Mg                    178 non-null    int64
 6   Phenols               178 non-null    float64
 7   Flavanoids            178 non-null    float64
 8   Nonflavanoid.phenols  178 non-null    float64
 9   Proanth               178 non-null    float64
 10  Color.int             178 non-null    float64
 11  Hue                   178 non-null    float64
 12  OD                    178 non-null    float64
 13  Proline               178 non-null    int64
dtypes: float64(11), int64(3)
memory usage: 19.6 KB
None
```

```
print(df.isnull().sum())
```

```
Wine                  0
Alcohol               0
Malic.acid            0
Ash                   0
Acl                   0
Mg                    0
Phenols               0
Flavanoids            0
Nonflavanoid.phenols  0
Proanth               0
Color.int             0
Hue                   0
OD                    0
Proline               0
dtype: int64
```

```
x = df.drop("Wine", axis=1)
y = df['Wine']
```

```
x_train, x_test, y_train, y_test, = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(x_train, y_train)
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (sta
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred = model.predict(x_test)
```

```
print("\nAccuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 1.0
```

```
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
Confusion Matrix:
[[14  0  0]
 [ 0 14  0]
 [ 0  0  8]]
```

```
print("\nClassification report:")
print(classification_report(y_test, y_pred))
```

```
Classification report:
              precision    recall  f1-score   support

           1       1.00      1.00      1.00        14
           2       1.00      1.00      1.00        14
           3       1.00      1.00      1.00         8

    accuracy                           1.00        36
   macro avg       1.00      1.00      1.00        36
weighted avg       1.00      1.00      1.00        36
```